Voice Activity Detection

Energy VAD vs. Lightweight CNN — Stability Under Noise

github.com/LaSibyl/vad-project

Introduction

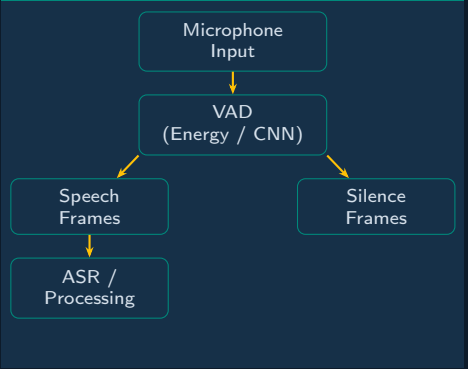
What is Voice Activity Detection?

- Segment a continuous audio stream into speech and non-speech frames at 10 ms resolution
- Binary frame-level classification problem
- Operates in real time — latency matters

Why does it matter?

- Essential front-end for ASR, speaker diarization, and noise suppression (Zoom, Siri, Teams)
- Unstable VAD ⇒ entire downstream pipeline degrades
- Simple to define, surprisingly hard to generalise

Typical VAD Pipeline



Problem Being Studied

Two candidate approaches:

- **Energy VAD** — classical rule-based, zero training cost, deployable anywhere
- **CNN VAD** — data-driven, lightweight (<500 K params), learns spectral features

The open question:

Real environments have varying noise levels. Which approach stays reliable when SNR drops? Is the extra complexity of CNN worth it?

Research Question

Under varying noise conditions (SNR 0–20 dB), does a lightweight CNN VAD offer meaningfully better **stability** than classical Energy VAD — and at what practical cost?

Primary Metrics

- F1 variance across SNR levels
- Worst-case F1 (at 0 dB SNR)
- Frame-level inference latency

Datasets

Final datasets used:

Dataset	Role
LibriSpeech train-clean-5	Training speech ($\sim$5 h)
LibriSpeech dev-clean-2	Validation
MUSAN noise	Noise augmentation + negative samples
LibriSpeech pause segments (RMS < 0.005)	Real silence negatives
`sample.wav` (6.21 s, 16 kHz)	Evaluation (manually labelled)

SNR sweep: MUSAN noise mixed at $\{0, 5, 10, 15, 20\}$ dB into `sample.wav`.

We didn't start here...

Our first training set was `SyntheticVADDataset` — a simplified synthetic dataset:

- **Speech:** random sinusoids (80–300 Hz) + small Gaussian noise
- **Non-speech:** low-amplitude white noise

Validation F1 = 0.92 $\longrightarrow$ Real-audio F1 = 0.05

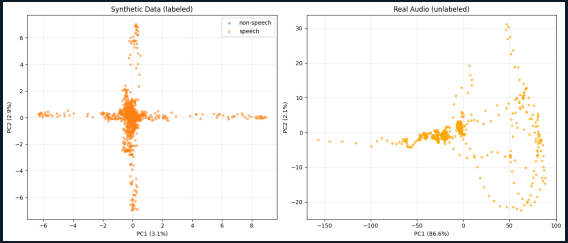
This mismatch drove every debugging decision that followed.

Tell a Story: Problem 1 — The Synthetic Trap

"Our model had never heard a real human voice."

Why `SyntheticVADDataset` failed:

Statistic	Synthetic	Real audio
Feature mean	≈ 0	$+5$ units
Feature std	0.5	1.2+
Feature range	2.76	> 10



Synthetic (top) vs. real-audio (bottom) feature distributions

The model's decision boundary lived in a completely different region of feature space from real audio — it had no chance of generalising to unseen recordings.

Fix: Switched training data to LibriSpeech + MUSAN real recordings; real-audio F1 immediately jumped from 0.05 to 0.42.

Tell a Story: Problem 2 — Real Data, Still Wrong

After switching to LibriSpeech + MUSAN, two independent failures appeared:

Problem A: Predicted Almost All Zero

- Training: `librosa` mel filterbank
- Inference: `numpy` FFT pipeline
- Different implementations $\Rightarrow$ 5-unit feature mean shift
- Model's threshold in wrong place — everything classified as silence

Fix: Unified `numpy` FFT for both training & inference

F1: 0.05 $\xrightarrow{\text{fix}}$ 0.42

Problem B: FAR = 0.96

- Training negatives: MUSAN noise (std $\approx$ 0.89)
- Real recording silence: std $\approx$ 0.44, different spectral shape
- Model had never seen real silence — misclassified it as speech

Fix: Mine LibriSpeech pause segments (RMS $<$ 0.005) as real silence negatives

F1: 0.42 $\xrightarrow{\text{fix}}$ 0.932, FAR: 0.96 $\rightarrow$ 0.002

Both failures share the same root cause: distribution mismatch between training and deployment.

Three Numbers Tell the Story

$$0.05 \longrightarrow 0.42 \longrightarrow 0.932$$

Real-audio F1 after each fix

Lesson 1

Synthetic data $\neq$ real audio distribution. Val F1 cannot be trusted without real-world testing.

Lesson 2

Training and inference **must use the identical feature pipeline**. Any library difference = domain shift.

Lesson 3

Your negative class must reflect **real negative examples**. Synthetic substitutes generalize poorly.

Now: how do the two VAD methods work, and how do they compare across SNR?

Methodology: Energy VAD

Raw Signal → Framing (**25 ms / 10 ms hop**) → Log Energy ($E_t = \log(\sum x_t^2 + \varepsilon)$) → Threshold ($\theta = \bar{E}$ **(adaptive)**) → Median Filter (**kernel size = 5**) → **Speech / Non-speech**

Key properties:

- Threshold is **adaptive per recording**: $\theta = \bar{E}$ (mean log-energy) — naturally adjusts to overall level
- Median filter suppresses isolated spurious frames
- **No training required** — rule-based decision

Cost

- Latency: 0.005 ms/frame
- Dependencies: `numpy` only
- Parameters: 0

Weakness: one energy threshold cannot distinguish spectrally complex noise from speech.

Methodology: CNN VAD

Raw Signal → Log-Mel Features (40 bins × 5 frames = 200-dim) → Z-score Norm (saved scaler) → LightweightCNN (1D Conv + BN + FC, 51,969 params) → Sigmoid ($\theta = 0.5$) → **Speech / Non-speech**

Training data (4 categories, 25 % each):

Category	Source
Speech only	LibriSpeech train-clean-5
Noise only	MUSAN noise
Speech + noise	Mix at SNR 5–20 dB
Real silence	LibriSpeech pauses (RMS < 0.005)

Cost

- Latency: 0.34 ms/frame (CPU)
- Loss: `BCEWithLogitsLoss`
- 20 epochs, global z-score scaler
- Requires labelled real data

List of Investigations Accomplished

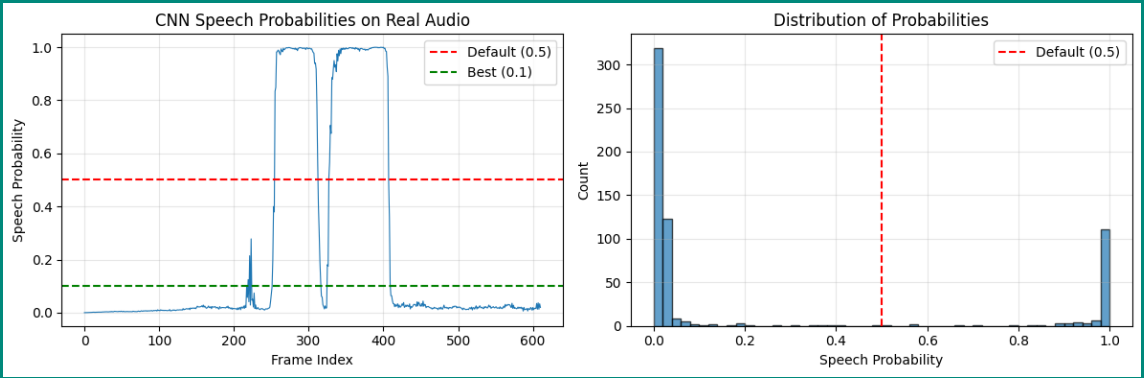
#	Symptom	Root Cause	Fix	Real F1
1	All probs $\rightarrow 1$	Double `sigmoid` + wrong `pos_weight`	Switch to `BCEWithLogitsLoss`	val 0.92
2	F1 = 0.05 on real audio	`librosa` vs `numpy`: 5-unit feature shift	Unified `numpy` FFT pipeline	0.42
3	FAR = 0.96 (silence = speech)	Negatives = MUSAN only; real silence std 0.44 vs 0.89	Mine LibriSpeech pauses (RMS < 0.005)	**0.932**

Also completed

- Global z-score scaler (train split)
- 4 training strategies (none / uniform / curriculum / real_data)
- Threshold sweep (0.1–0.9)
- SNR stability sweep (5 levels)

Each failure was a manifestation of the same root cause: distribution mismatch between training data and real deployment environment. This is the central challenge in applied ML — not model architecture, but data alignment.

Results: Single Recording (`sample.wav`)

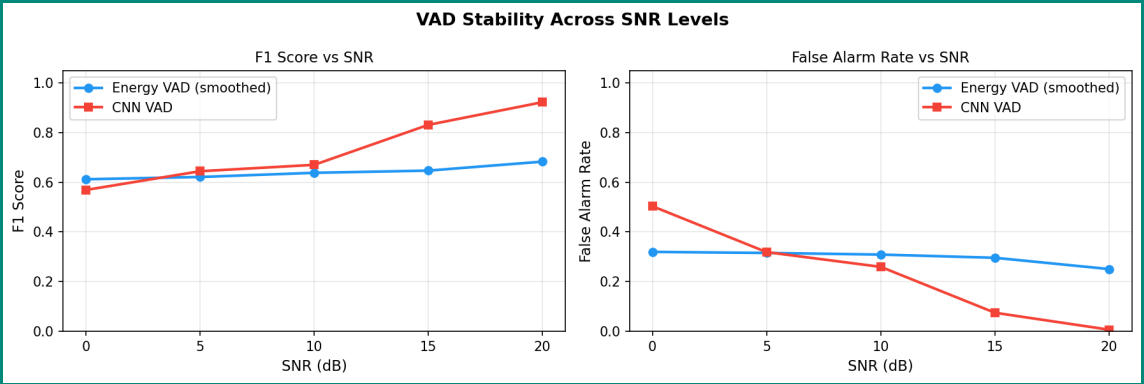


Method	F1	FAR	MISS
Energy (raw)	0.899	0.039	0.090
Energy (smooth)	0.922	0.022	0.090
CNN (thr = 0.5)	0.932	0.002	0.122
CNN (thr = 0.1)	**0.945**	0.015	0.064

Key Findings

- CNN FAR: 11× lower than smoothed Energy VAD
- Threshold tuning trades FAR vs MISS rate

Main Results: SNR Stability (0–20 dB)



SNR	E-F1	E-FAR	C-F1	C-FAR
0	0.612	0.320	0.569	0.503
5	0.621	0.315	0.644	0.319
10	0.638	0.309	0.670	0.259
15	0.647	0.296	0.831	0.075
20	0.683	0.251	**0.922**	**0.007**

Key Findings

- Crossover ≈10 dB: Energy wins below, CNN above
- F1 std: Energy 0.025 vs CNN 0.130 (5× more stable)

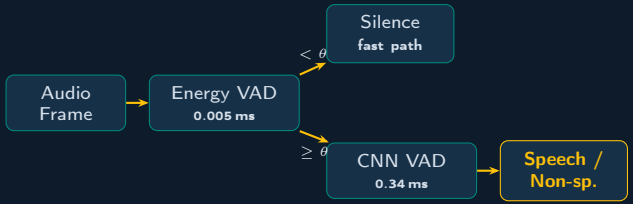
Latency & Practical Trade-off

Frame-level latency (CPU):

Method	ms/frame	vs Energy
Energy VAD	0.005	baseline
CNN VAD	0.340	71× slower

At 10 ms hop: Energy uses ≈0.05 % real-time budget; CNN uses ≈3.4 %. Both viable for streaming.

Cascade design (future work):



CNN load reduced ∼70 %

Application-level decision matrix:

Scenario	Recommended
Edge / embedded (no PyTorch)	Energy VAD
SNR < 10 dB (very noisy)	Energy VAD
No labelled training data	Energy VAD
Strict latency (<0.01 ms/frame)	Energy VAD
Office / phone (SNR ≥ 10 dB)	CNN VAD
Strict false-alarm requirement	CNN VAD
System already runs PyTorch	CNN VAD

What We Learned / Would Do Differently

Key Takeaways

1. **Distribution shift is subtle.** Synthetic data can produce a false sense of validation performance.
2. **Feature pipelines must be identical** end-to-end. Any library difference = domain shift.
3. **Negative class matters as much as positive.** MUSAN noise $\neq$ real silence.
4. **CNN $\neq$ universally more stable.** Stability depends on SNR regime; Energy VAD wins below 10 dB.

Would Do Differently

- Evaluate on **multiple recordings**, not just one `sample.wav` (low statistical confidence with $n = 1$)
- Extend training SNR range to <0 dB to improve low-SNR robustness
- Implement **cascade** as primary architecture (Energy gate + CNN refine)
- Curriculum training: high-SNR first, then progressively harder
- Add Room Impulse Responses (RIRS_NOISES) to training

Conclusion

Answering the Research Question

CNN VAD is **not** uniformly more stable. Stability is SNR-dependent:

- <10 dB: Energy VAD is more reliable (lower variance, no collapse at 0 dB)
- ≥ 10 dB: CNN VAD is dramatically better (F1 $+35\,\%$, FAR $\div 11$ at 20 dB)

The practically optimal architecture is a cascade of both methods.

Metric	Energy VAD	CNN VAD
F1 mean (0–20 dB)	0.640	**0.727**
F1 std	**0.025**	0.130
Peak F1 (20 dB)	0.683	**0.922**
Worst F1 (0 dB)	**0.612**	0.569
Latency	**0.005 ms**	0.340 ms

What We Built

- Full CNN VAD pipeline: data $\rightarrow$ training $\rightarrow$ deployment
- 3-round systematic debugging ($0.05 \rightarrow 0.932$)
- SNR stability sweep across 5 noise levels
- Shared evaluation module (F1, FAR, MISS)
- LaTeX reports, updated README, reproducible code

Thank you!

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